

Belief-State Geometry in Transformers: Reproduction and Unsupervised Recovery via Observable Operator Models

1 Introduction

Shai et al. [1] show that a transformer trained on sequences from a hidden data-generating process linearly represents, in its residual stream, the geometry of *belief states*: the Bayesian posteriors over the generator’s hidden state given the observed history. The relevant object is the *mixed-state presentation* (MSP) from computational mechanics [7, 8]: each history w induces a belief $b(w) \in \Delta^{n-1}$ over the n hidden states, and the set $\{b(w)\}$ forms a characteristic fractal (or finite point set) in the simplex. An optimally predicting network must, and does, encode this geometry affinely in its activations.

Section A reproduces the central supervised results for two processes, Mess3 and RRXOR. Section B is a contribution: an *unsupervised* estimator that recovers the belief-state geometry from activations and the network’s own next-token distribution alone — using no ground-truth belief labels and no knowledge of the generating process — built on the theory of *observable operator models* (OOMs) [3, 4], presented side by side with the supervised representation. The Appendix studies the generative *operator rollout*.

Processes and model. *Mess3* ($x=0.05, a=0.85$) is a 3-state, vocab-3 process whose MSP is a dense Sierpinski-like fractal. *RRXOR* is a 5-state, vocab-2 synchronising process whose MSP is a finite set of 36 belief states. We train the paper’s architecture (App. A.6: $d_{\text{model}}=64$, 4 layers, 1 head, $d_{\text{mlp}}=256$, ReLU, LayerNorm, SGD) for 10^6 steps. Both reach their myopic-optimal loss (Mess3 ≈ 0.7935 ; RRXOR ≈ 0.5788). Throughout, regressions and operator fits use *prefix deduplication* (causal attention makes a prefix’s activation identical at every occurrence) and $P(w)$ -*weighting* by prefix probability, following the convention of [2].

2 Section A: Supervised reproduction

For each process we enumerate every positive-probability length-10 input, take the ground-truth belief after each position, regress the (concatenated, per-layer) residual stream onto the belief, and project the predictions with the same PCA as the ground-truth simplex.

2.1 Mess3

The belief geometry appears in the final-layer residual stream as the characteristic fractal (Figure 1, belief-decode $R^2 \approx 0.99$). It is not present at initialisation but *emerges* over training, and is non-trivial: it survives cross-validation on held-out prefixes (so it is not memorisation) and collapses to the mean under a belief-label shuffle (so it is not an artefact of projecting onto a fractal-shaped target). Figure 2 reproduces the paper’s Fig. 6 in full.

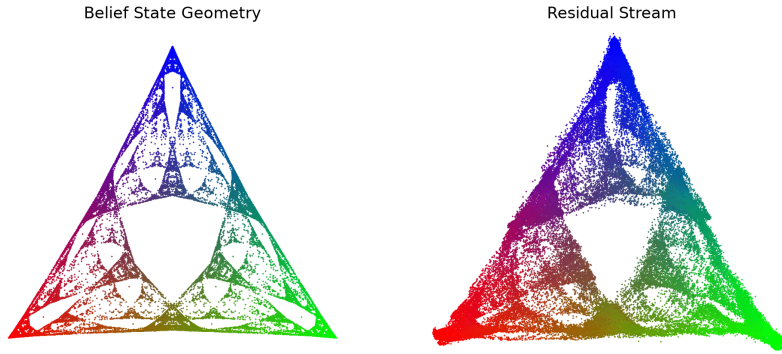


Figure 1: **Mess3, paper Fig. 5.** Belief geometry recovered from the final-layer residual stream by linear regression onto belief and projected to the simplex (datashader-style raster).

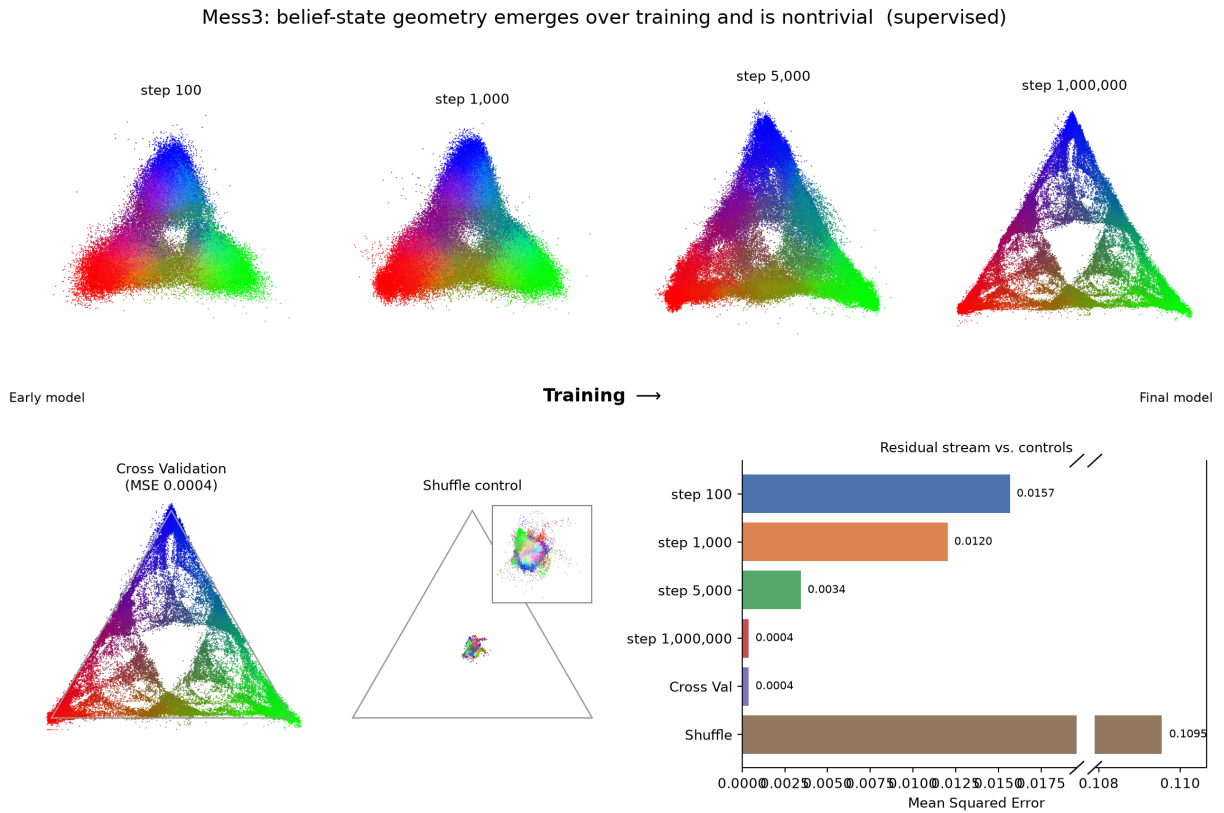


Figure 2: **Mess3, paper Fig. 6 (supervised).** *Top:* belief-fractal emergence across training checkpoints (steps 100 to 10^6). *Bottom:* cross-validation (the fractal persists on held-out prefixes), shuffle control (decode collapses to the mean), and belief-reconstruction MSE at each checkpoint plus cross-validation and shuffle on the final model.

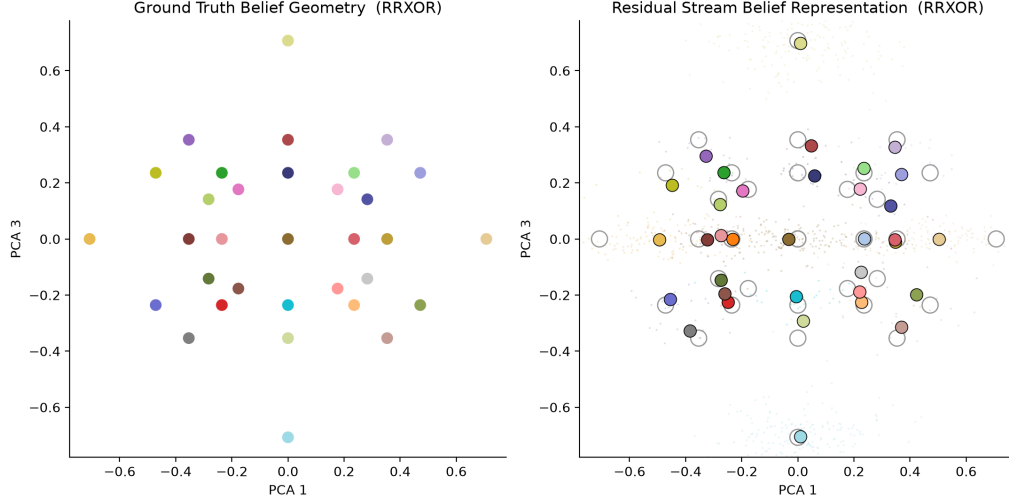


Figure 3: **RRXOR**, paper Fig. 7C. Ground truth (left) and supervised recovery from the concatenated residual stream (right; prefix-deduplicated, $P(w)$ -weighted; small dots = per-unique-prefix predictions, large = per-state COM, grey rings = ground truth).

2.2 RRXOR

RRXOR’s belief geometry is the symmetric four-apex diamond of the 36 MSP states in the PC1/PC3 plane (left panel of Figure 3). It is recovered from the concatenated residual stream (right panel) at $R^2 \approx 0.87$ (prefix-deduplicated, $P(w)$ -weighted), with the 36 state centres of mass at $R^2 = 1.00$. Two further controls match the paper (Figure 4): the geometry is spread across *all* layers (every single layer carries it poorly, the concatenation best), and belief distances are preserved ($R^2 \approx 0.78$) while next-token distances are not ($R^2 \approx 0.31$) — the geometry is not a next-token readout.

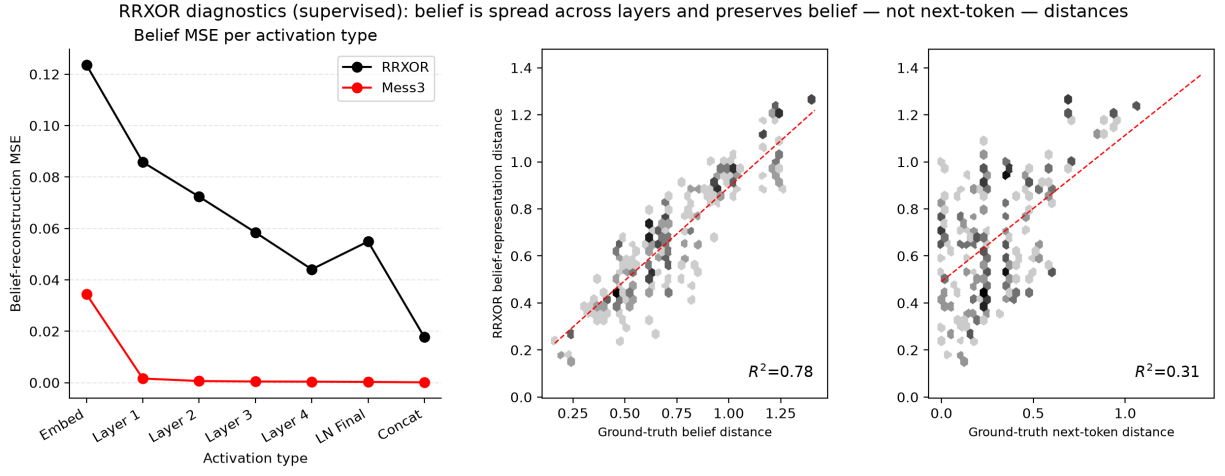


Figure 4: **RRXOR diagnostics, supervised (paper Fig. 7D–E).** *Left:* per-activation belief-reconstruction MSE — RRXOR’s geometry is spread across all layers (every single layer high; the concatenation lowest), while Mess3 carries it in every layer. *Centre/right:* over the 36 belief states, the representation preserves belief distances ($R^2 \approx 0.78$) but not next-token distances ($R^2 \approx 0.31$) — the geometry is more than next-token.

3 Section B: Supervised vs. unsupervised recovery

Observable operator models. A discrete stochastic process over an alphabet Σ admits an OOM [3]: a state space $\mathbb{R}^d$, one *observable operator* $\tau_x \in \mathbb{R}^{d \times d}$ per symbol, an initial state π , and an evaluation functional σ , with every word probability

$$\Pr(x_1 \cdots x_L) = \sigma \tau_{x_L} \cdots \tau_{x_1} \pi. \quad (1)$$

Hidden Markov models are the special case $\tau_x = T^x$, $\sigma = \mathbf{1}^\top$, with $T_{ij}^x = \Pr(\text{state } j, \text{emit } x \mid \text{state } i)$. The Bayesian *belief* $b(w)$ is the OOM state after w , updated projectively

$$b(wx) = \frac{b(w) T^x}{b(w) T^x \mathbf{1}}, \quad \Pr(x \mid w) = b(w) T^x \mathbf{1}. \quad (2)$$

OOMs, predictive-state representations [6] and multiplicity automata are unified by Thon and Jaeger [4]; the subspace-learning idea below is the one used by Hsu, Kakade and Zhang for HMMs [5]: a process of OOM-dimension d is recovered, up to a similarity (gauge) transform, from the rank- d factorisation of a past×future statistic.

The observability-OOM estimator. Section A establishes that the network encodes belief affinely, $a(w) = b(w)\Psi$. The projective update (2) is non-linear in a , but multiplying through by the normaliser — which the network *outputs* as its softmax $\Pr(x \mid w)$ — linearises it: $a(w)A_x \approx \Pr(x \mid w) a(wx)$ with $A_x = \Psi^+ T^x \Psi$. Let $C : a(w) \mapsto \Pr(\cdot \mid w)$ be the one-step observable and $\{G_x\}$ these rescaled operators (all $P(w)$ -weighted ridge regressions). The *observability matrix* $\mathcal{O} = [C \ G_x C \ G_x G_y C \ \cdots]$ stacks the multi-step observables, each linear in belief; an SVD of $\mathcal{O}$ returns an orthonormal basis B , with the rank giving the model order d . Everything uses only activations, the model softmax and the prefix tree; ground-truth belief is used solely to *score*. The construction assumes only linearity, so it transfers unchanged to quantum/post-quantum representations.

The figures below are strict like-for-like: identical pipeline, identical PCA, identical (label-fit) placement onto the simplex; the *only* difference between the supervised and unsupervised panels is what feeds the affine readout — the full residual stream versus the recovered d -dimensional observability subspace.

3.1 Mess3

The observability spectrum drops cleanly at $d=3$, $\text{eig}(A_x)$ matches $\text{eig}(T^x)$, and the recovered 3-D subspace decodes belief at $R^2 = 0.998$ — indistinguishable from the supervised 0.999 (Figure 5). Mess3 is strongly observable, so the labels add essentially nothing.

The agreement holds over training and under the same controls as Section A, with the identical observability-OOM + $P(w)$ estimator. The belief fractal emerges in lockstep in the supervised and unsupervised representations across checkpoints (Figure 6), and both survive cross-validation and collapse under shuffle at comparable reconstruction error (Figure 7). Both are cross-validated on a stringent 30/70 split (train 30%, test 70%). This is the lowest train fraction the unsupervised method tolerates: its training unit is a *transition* (a prefix together with all its children), which survives a random prefix split at only $\sim f^4$, so below $\sim 30\%$ train it runs out of transitions to estimate the subspace; the supervised decode, whose unit is a single prefix, is unaffected.

3.2 RRXOR

For RRXOR the unsupervised subspace recovers the diamond but more diffusely (Figure 8): per-(input,position) $R^2 = 0.65$ versus the supervised 0.91, and state-COM geometry $R^2 \approx 0.82$ versus

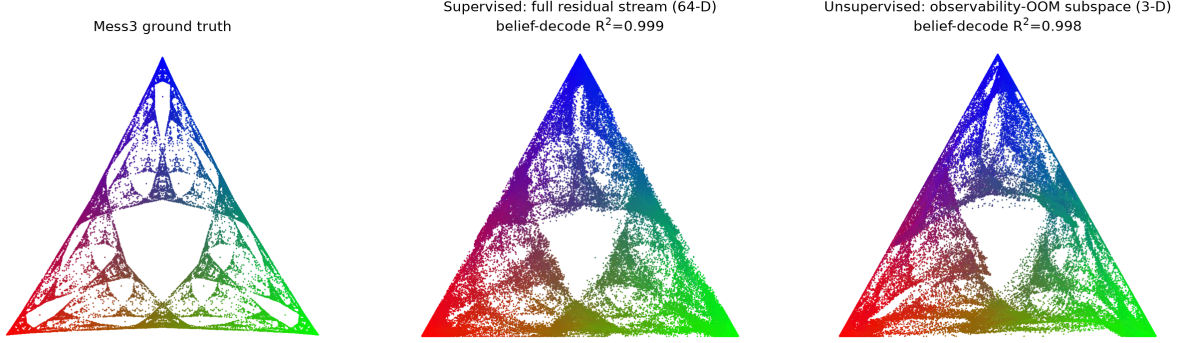


Figure 5: **Mess3, supervised vs. unsupervised.** Ground truth (left); supervised full-residual decode (centre, $R^2 = 0.999$); unsupervised observability-OOM subspace (right, $R^2 = 0.998$). Identical pipeline; only the affine-map input differs.

1.00. The gap is genuine, not an estimator artefact: several independent unsupervised methods converge to a similar state-level ceiling, the observability spectrum has no clean rank-5 elbow, and a control confirms the limit is structural — PCA of the 36 true state-COMs recovers only $R^2 = 0.35$, so belief is a *low-variance* feature even among the state centres, and only predictive (not variance-based) subspaces work. The cause is *weak observability*: RRXOR’s belief has directions that barely change the predicted future (the multi-step version of the 0.33 next-token result), surfacing only as faint singular values that need a deep horizon — which collides with the $n_{\text{ctx}}=10$ coverage wall. The supervised regression sidesteps this with labels; the gap is the “value of the labels,” near zero for the strongly observable Mess3 and large for RRXOR.

The two RRXOR controls also have unsupervised counterparts (Figure 9). The “belief spread across layers” result is recovered without labels: each single layer’s observability-OOM subspace decodes belief poorly while the concatenation does best (left panel, mirroring Figure 4). The distance control, however, *reverses*: the recovered representation preserves next-token distances ($R^2 \approx 0.68$) more than belief distances ($R^2 \approx 0.50$), the opposite of the supervised case. This is consistent with the rest of the section rather than a contradiction — the observability-OOM subspace is anchored on the next-token observable C , so its dominant geometry is next-token-shaped, and belief is a low-variance feature within it: linearly recoverable by the readout of Figure 8, but not the leading distance structure. It is a direct manifestation of RRXOR’s weak observability.

4 Conclusion

The supervised results of Shai et al. [1] reproduce cleanly for both processes, and the belief-state geometry can moreover be recovered *unsupervised*, from activations and the model’s own softmax, using a classical observable-operator construction: essentially perfectly for the strongly observable Mess3, and partially — to a quantifiable ceiling set by observability — for RRXOR. The determining factor is process observability, not the classical-vs-post-quantum distinction, which suggests the natural next experiment is a richly sampled post-quantum process, where both subspace and operators should come out clean.

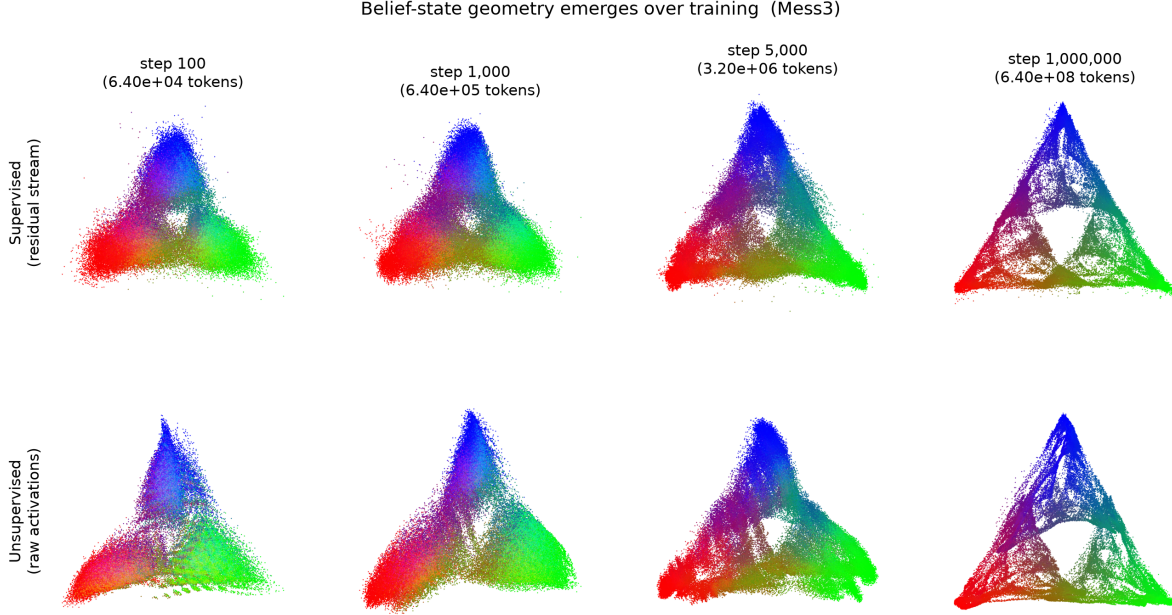


Figure 6: **Mess3 emergence, supervised vs. unsupervised.** Belief-fractal emergence across training checkpoints: top row = supervised residual-stream decode; bottom row = unsupervised recovery from the raw activations. The geometry sharpens in lockstep in both.

A Operator rollouts

The subspace recovery above tests “is belief linearly present?” A stronger test is the generative *operator rollout*: discard the per-prefix activations and regenerate every belief state purely from the learned operators (seeded only at the root, $s(wx) = s(w)A_x / s(w)A_x e$, with e the recovered evaluation functional). We learn the operators by the same $P(w)$ -weighted rescaled least squares, with an ALS operator-consistency refinement, and apply the identical recipe to the supervised subspace (Figures 10, 12) and the unsupervised one (Figures 11, 13).

For **Mess3** the rollout reproduces the full fractal at $R^2 = 1.00$ from *both* subspaces: its operators are diagonalisable strict contractions whose attractor *is* the belief geometry. For **RRXOR** the rollout degrades and does not reach the outer apexes from either subspace (the COM-geom numbers, ~ 0.6 – 0.7 , are flattered by the label-fit readout; visually the extreme states stay near the centre). This is intrinsic to RRXOR’s operator algebra — its true belief-update operators are non-diagonalisable (T^1 nilpotent, T^0 defective with a triple eigenvalue), so the belief states are transients of a defective semigroup, not the attractor of a contraction, and iterating estimated operators cannot recover the weakly-observable extreme states for *any* subspace. Hence for RRXOR the raw subspace (Section B), not the rollout, is the trustworthy representation.

References

- [1] A. S. Shai, S. E. Marzen, L. Teixeira, A. G. Oldenziel, P. M. Riechers. *Transformers represent belief state geometry in their residual stream*. NeurIPS 2024; arXiv:2405.15943.
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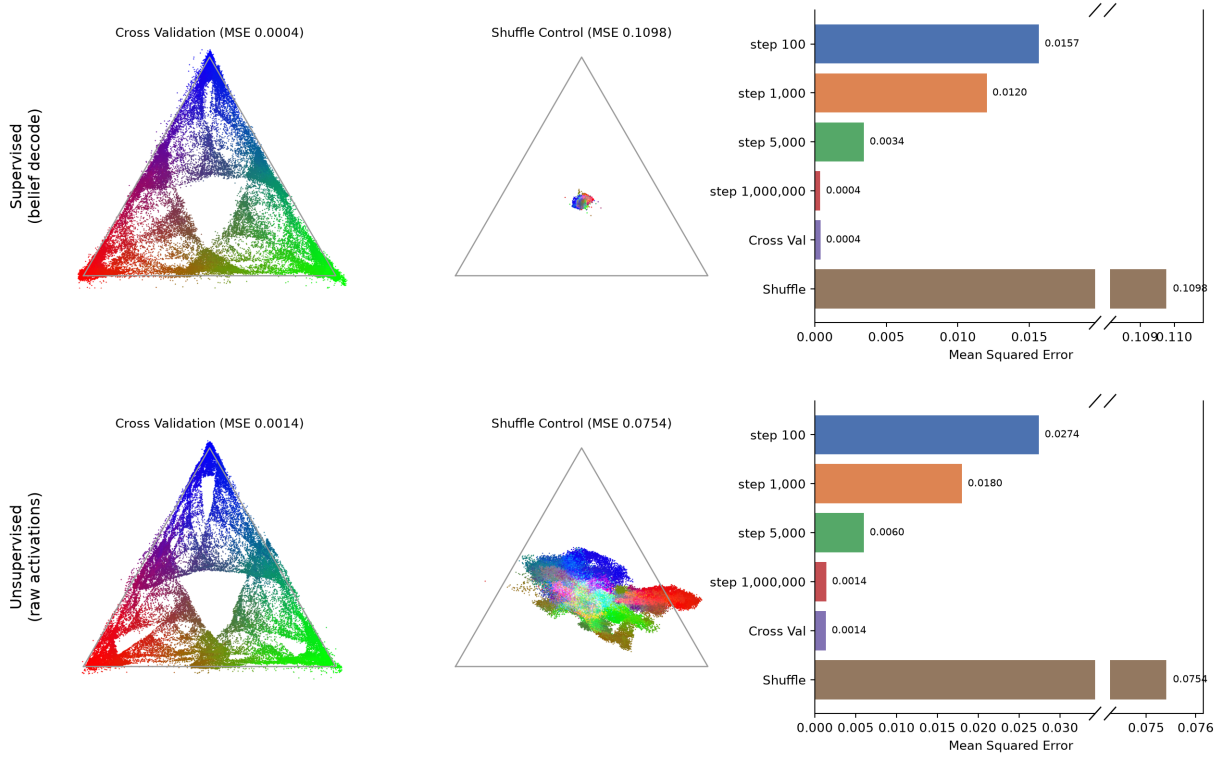


Figure 7: **Mess3 controls, supervised vs. unsupervised.** Cross-validation, shuffle control, and per-checkpoint / CV / shuffle MSE bars, for the supervised (top) and unsupervised (bottom) representations. Both recover the fractal on held-out prefixes and collapse under shuffle.

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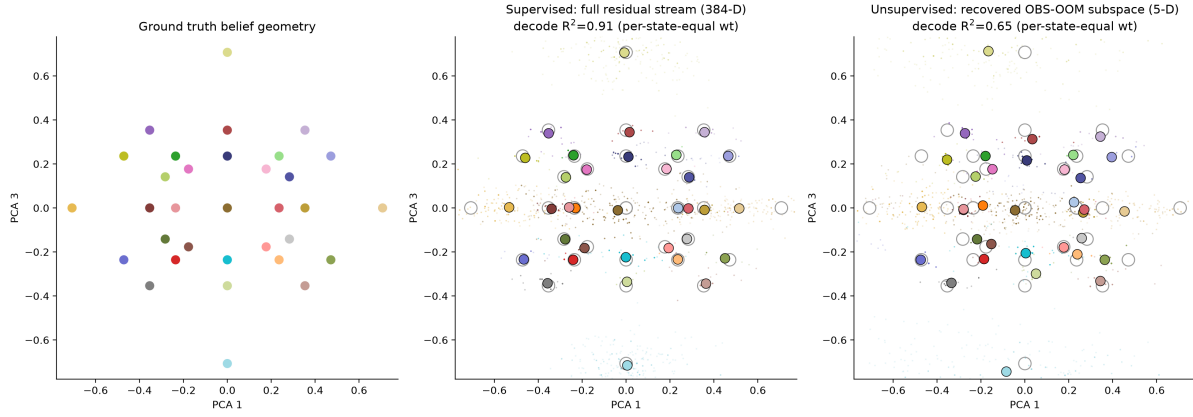


Figure 8: **RRXOR, supervised vs. unsupervised.** Ground truth (left); supervised full-residual recovery (centre, per-position $R^2 = 0.91$); unsupervised 5-D observability-OOM subspace (right, per-position $R^2 = 0.65$, state-COM $R^2 \approx 0.82$). Identical pipeline; only the affine-map input differs. Placement onto the simplex uses a label-fit readout in *both* panels (display only — the unsupervised content is the subspace).

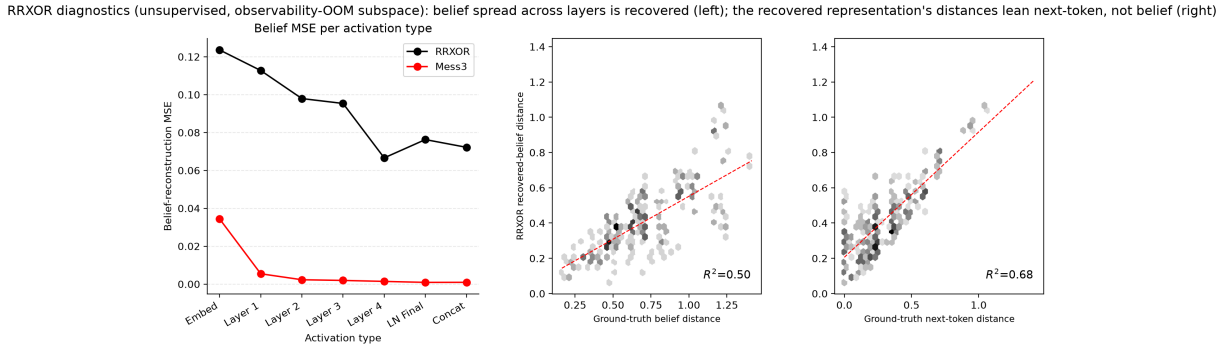


Figure 9: **RRXOR diagnostics, unsupervised counterpart.** *Left:* per-activation belief-decode MSE from each activation’s observability-OOM subspace — “belief spread across layers” is recovered without labels (cf. Figure 4). *Centre/right:* distance preservation of the recovered representation — unlike the supervised case it leans toward next-token ($R^2 \approx 0.68$) over belief ($R^2 \approx 0.50$), because the observability subspace is anchored on the next-token observable.

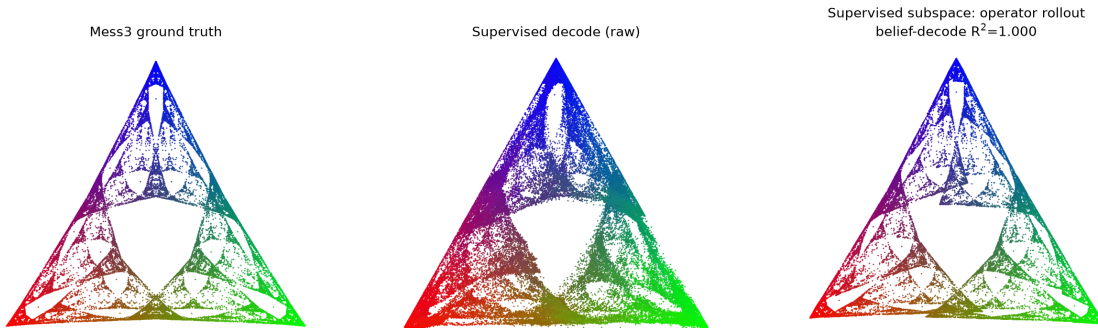


Figure 10: **Mess3, supervised subspace.** Ground truth — supervised decode — operator rollout from the supervised subspace ($R^2 = 1.00$).

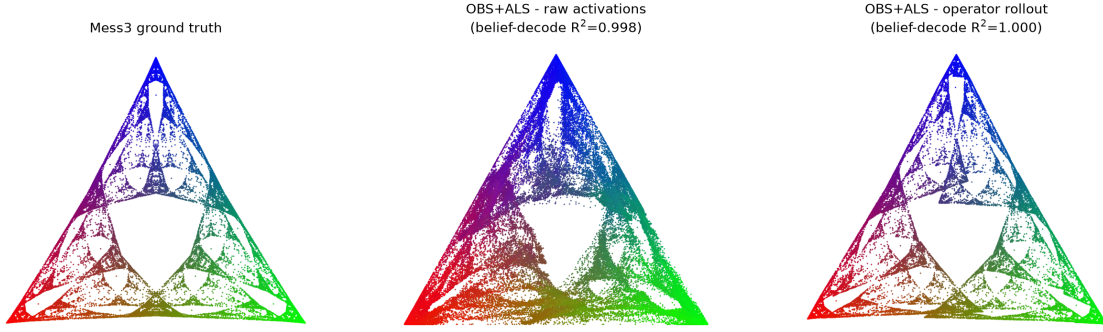


Figure 11: **Mess3, unsupervised subspace.** Ground truth — unsupervised observability-OOM subspace — operator rollout ($R^2 = 1.00$).

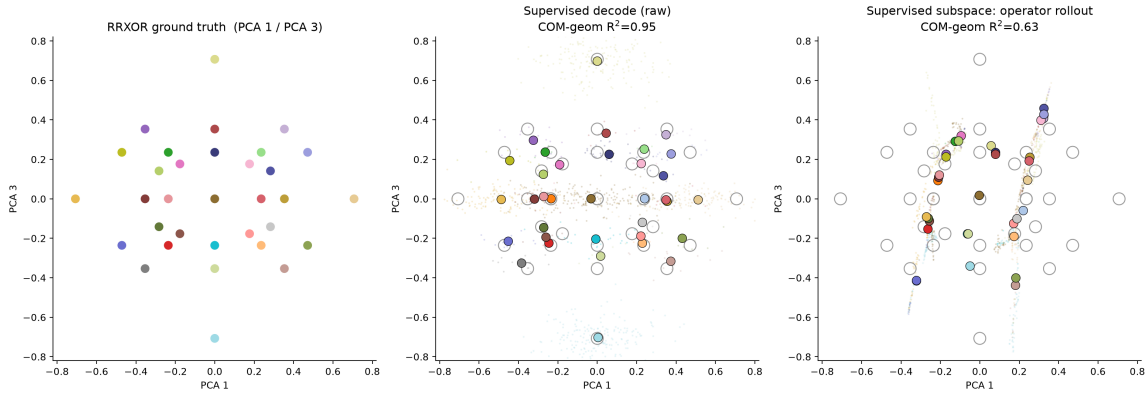


Figure 12: **RRXOR, supervised subspace.** Ground truth — supervised decode (COM-geom 0.95) — operator rollout (COM-geom 0.63; the extreme states are not reached).

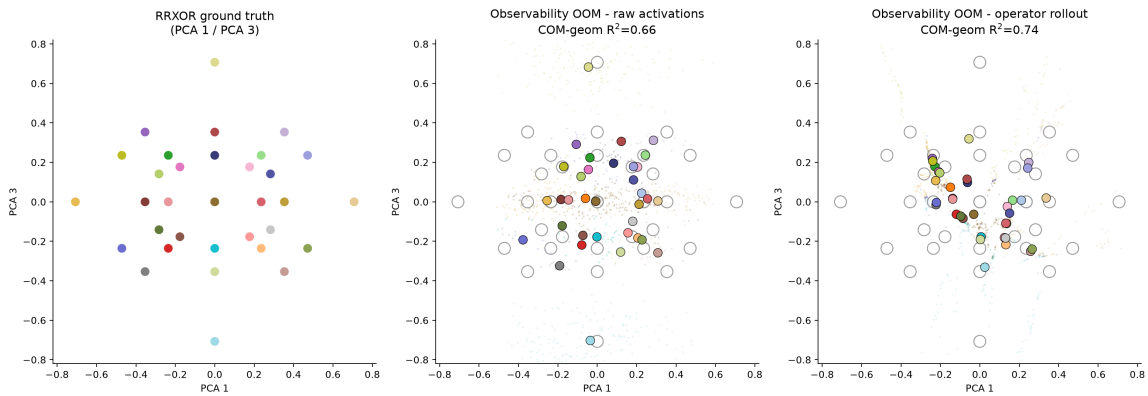


Figure 13: **RRXOR, unsupervised subspace.** Ground truth — unsupervised observability-OOM subspace (COM-geom 0.66) — operator rollout (COM-geom 0.74; metric flattered, the extreme states stay central).